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YASH PATEL,¹ PRAJAKTA BELEKAR,¹ GEORG LADURNER,¹ SYBREN WORM,¹ LUCAS MAY,¹ MARIA VARAKA,¹ THERESA ROHRMOSER,¹ FERNANDO GARCÍA-RAMÍREZ,¹ RENÉ WERKMEISTER,¹ BERNHARD BAUMANN,^{1,2} AND CONRAD W. MERKLE^{1,*}

¹Center for Medical Physics and Biomedical Engineering, Medical University of Vienna, Währinger Gürtel 18-20/4L, 1090 Vienna, Austria

²Institute of Biomedical Physics, Medical University of Innsbruck, Müllerstraße 44, 6020 Innsbruck, Austria

*conrad.merkle@meduniwien.ac.at

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1. Introduction

General structure of intro: Why diameter is an important measurment Brief OCTA Introduction Difficulty of OCTA quantification (different methods yield different results) Artefact contributions and diameter underestimation Proposed solution that we investigate here and evaluation of different methods in a test scenario (vessel dilation)

<https://www.nature.com/articles/s41598-021-84067-2> <https://qims.amegroups.org/article/view/51210/html>

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Vessel diameter is an important biomarker that can provide valuable information regarding local vascular health. For example, studies have demonstrated that vessel lumen diameter in the retina decreases with increasing severity of diabetic retinopathy [cite]. Combined with measurements of blood flow, diameter can also be used to investigate local blood pressure and vessel elasticity [cite]. For this reason, it is imperative that accurate measurements of diameter be used for these extrapolations or when modeling behaviours of flow in digital twins [cite]. Optical imaging techniques are often used to measure vessel diameter due to their high resolution, but accurate measurement of vessel size is more difficult than it first appears.

Optical coherence tomography (OCT) is a non-invasive optical imaging method, which uses interferometry to measure the transit time of light that is backscattered from a sample, thereby producing depth-resolved images of tissue [?, ?]. Optical coherence tomography angiography (OCTA) is an extension of OCT that visualizes vascular perfusion using the motion contrast from red blood cells (RBCs) [?, ?, ?, 1]. Over the years, OCTA has demonstrated widespread acceptance and use for clinical evaluations of vascular features in relation to disease [2]. OCT angiograms can be used to evaluate characteristics on the scale of vascular layers by measuring vascular density or non-perfusion zones [?], or it can be used to guage vessel size [?] or tortuosity [?].

Analysis of OCTA datasets is typically achieved by performing a maximum intensity projection of a 3D vascular network to compress it into 2D before binarising the resulting angiogram using a global or local binarisation method. From this binarised angiogram, various vascular metrics

can then be quantified. Importantly, there are many methods for performing this binarisation step and the choice of method can significantly affect the reported vascular metrics, with no clear consensus for a one size fits all approach [?, ?, ?, ?, ?]. Similarly, there are vascular enhancement algorithms to highlight structures that resemble vessels before quantification, but these methods also risk affecting the apparent size of the vessels in the network [?, ?, ?].

Furthermore, OCT is prone to certain artifacts which specifically affect visualization of the vasculature [?, 3, 4]. When viewed in cross-section, large vessels often demonstrate an hourglass appearance with stronger scattering in the center than at the sides (Figure 1A). This appearance is due to orientation effects of RBCs, which align with the blood flow, causing a reduced scattering cross-section at the sides of the vessels [3]. We hypothesize that these artifacts may lead to underestimations of measurements of vessel diameter. To compensate for these artifacts, OCT contrast agents such as Intralipid have previously been used to increase intravascular intensity signals, particularly in regions where RBC scattering is diminished [?, ?, 4–6].

In this study we examine different methods of measuring vessel size from in vivo OCT scans of mouse retinal vasculature, and we use a contrast agent to obtain ground truth vessel diameter measurements for comparison. In addition to a traditional binarisation-based methods, we also employ model-based methods for quantifying vessel diameter. Traditional Gaussian models have been employed for vessel diameter measurements in techniques such as [insert] [cite] (based on other optical imaging methods) and OCT [?], but we expand this by also investigating the generalized Gaussian model. As part of this study, we measure how much diameter is underestimated from enface projections and evaluate how reliably it can be compensated through calibration. After identifying the best methods for quantifying diameter and compensating for underestimations, we finally evaluated a test scenario using flicker stimulation to induce a measureable change in diameter to evaluate the performance of different methods.

2. Methods and Materials

2.1. Imaging system

All imaging was performed using a custom-built polarisation-sensitive spectral-domain OCT ophthalmoscope, described in detail previously [7]. The system was centred at 840 nm with a spectral bandwidth of 100 nm at full width half maximum (FWHM), providing an axial resolution of approximately 5.1 μm in air and 3.8 μm in tissue (assuming a refractive index of 1.35). A sensitivity of 96 dB was achieved at an A-scan rate of 80 kHz and a beam diameter of 0.5 mm at the pupil plane (power at cornea = 2.85 mW) [8]. This system was employed to acquire volumetric OCT data from mouse retinas before, during, and after an intravascular contrast agent administration, as well as before and after functional stimulation of the mouse retina using a custom 502 nm light-emitting diode (LED) ring flashing at 10 Hz (power at cornea = 26.6 μW). This wavelength was selected to target the peak absorption of mouse rhodopsin (~500 nm), enabling predominantly rod-driven photoreceptor stimulation [9].

2.2. Experimental protocol

Three mouse models were used in this study: VLDLR knockout mice (B6;129S7-*Vldlr*^{*tm1Her*}/J, The Jackson Laboratory, Bar Harbor, USA), C57BL/6J mice (C57BL/6J, The Jackson Laboratory, Bar Harbor, USA), and SOD1 non-transgenic mice (strain details), with $n = 1, 1, 2$ mice per VLDLR, SOD1 and C57BL/6J group respectively. All imaging sessions were conducted under general anaesthesia using either isoflurane, medetomidine + midazolam + fentanyl + ketamine (MMFK) combination, or a ketamine + xylazine combination. For the VLDLR knockout and the SOD1 non-transgenic mice, isoflurane was induced at 4% in oxygen for 4 minutes and subsequently maintained at 2% in oxygen at a flow rate of 1.5–2 L/min for the duration of the imaging session. In cases where isoflurane alone did not provide adequate sedation, an

intraperitoneal injection of ketamine + xylazine cocktail (100 mg/kg ketamine, 6 mg/kg xylazine; 10 mL/kg body weight) was administered. For the C57BL/6J mice, isoflurane was induced at 4% in oxygen for 4 minutes and then MMFK (10 mL/kg body weight) was administered. To maintain core body temperature of the mice throughout imaging, they were positioned on a heating pad. Mouse pupils were dilated with one drop of tropicamide (5 mg/mL, Agepha Pharma s.r.o., Senec, Slovakia) per eye, and artificial tear drops (Oculotect, ALCON Pharma GmbH, Freiburg im Breisgau, Germany) were applied at regular intervals to prevent the cornea from drying out.

Intralipid 20% (3 mL/kg body weight) was used as a contrast agent and was administered as a bolus injection via the tail vein. Volumetric angiography data were acquired over a 1 mm × 1 mm field of view centred on the optic nerve head (ONH), with a scan density of 500 A-scans per B-scan and five repeated B-scans at each of 400 slow-axis positions (acquisition time approximately 16 seconds). For this study, pre- and post-injection volumetric datasets were acquired. **This volumetric data was used solely for visualisation purposes.** Additionally, a dynamic-contrast OCT (DyC-OCT) protocol [6] was employed during the contrast injection or during stimulation, consisting of 2000 repeated B-scans over approximately 16 seconds (interscan time 8.2 ms) at the same cross-sectional location over a 1 mm lateral field of view. DyC-OCT scans allow us to track the same vessels over time to precisely measure the effects of the contrast agent or the stimulation with a high degree of spatial coregistration. All quantitative vessel diameter measurements reported in this study were derived from the DyC-OCT data.

Functional flicker stimulation was performed in a C57BL/6 mouse to assess whether the choice of diameter measurement method affects the detection of physiological vascular responses. For this experiment, ketamine/xylazine cocktail was chosen over isoflurane to minimise anaesthesia-induced vasodilation [cite]. During the DyC-OCT acquisition, the retina was unstimulated for approximately 7 seconds, followed by 10 Hz flicker stimulation from the LED ring for the rest of the scan duration. **To compare results before and after stimulation, the data was divided into baseline (pre-stim) and post-stim. Considering the time required for the OCT signal to stabilise after stimulation, four-second time windows at the beginning and end of each scan were selected for the pre- and post-stimulation analyses respectively.**

All animal procedures were approved by the ethics committee of the Medical University of Vienna and the Austrian Federal Ministry of Education, Science, and Research (BMBWF/66.009/0272-V/3b/2019 and GZ 2024-0.802.524).

2.3. Post-processing and analysis

Cross-polarised OCT intensity data from the retinal pigment epithelium (RPE) was used for inter-frame alignment and B-scan flattening [8, 10]. Angiograms were derived by applying the complex subtraction OCT angiography (OCTA) algorithm [1, 11]. We evaluated multiple combinations of methods for quantifying vessel diameter from the OCTA data to determine which combinations of steps work best under different conditions. Specifically, we looked at two types of data projections (maximum intensity and mean intensity) to generate vessel profiles and several different quantifications of those profiles including a traditional binarisation, a Gaussian model approach, and a Generalised-Gaussian (GG) model approach. These methods represent commonly used methods in the literature for OCTA quantification [12–15]. The different methods were evaluated against the ground truth as described in this section. All data processing and analysis was performed in MATLAB (R2019b, MathWorks, Natick, MA, USA).

2.3.1. Ground truth calculation

Ground truth vessel diameters were obtained from contrast-enhanced DyC-OCT B-scan angiograms. The lateral (d_x) and axial (d_z) diameters of the vessel were manually measured from the vessel's cross-section. The ground truth diameter was defined as the minimum of the two measurements to account for off-axis cross-sectioning of the vessel, which produces a more

elliptical appearance in the B-scan. For a vessel perpendicular to the B-scan, the lateral and axial diameters would therefore be in agreement.

2.3.2. Vessel projection calculation

For each large vessel in the field of view, a rectangular region of interest (ROI) encompassing the vessel cross-section, along the lateral (x) and axial (z) dimensions, was selected from the DyC-OCT B-scan angiograms. En face intensity profiles were then generated by projecting the A-scans within this ROI along the axial (z) direction. Two projection methods were compared: maximum intensity projection (MIP), which selects the highest intensity value along the depth axis of each ROI, and mean intensity projection (MeIP), which averages the values along each ROI. Each projection resulted in a time-stack of 1-D lateral intensity profiles for each vessel, which were then used for subsequent diameter quantification. For the model-based approaches, the profiles were normalised to a range of 0–1 to ensure a good fit.

2.3.3. Skew correction

Off-axis cross-sectioning elongates the vessel along the lateral direction, so the lateral diameter measured from the ROI projection can overestimate the true vessel size. Because the lateral diameter is the quantity studied here, a correction factor of d_z/d_x was defined for cases where the lateral diameter exceeded the axial diameter, and was equal to 1 otherwise. By multiplying this correction factor against the diameter measured from the ROI projection, we ensured that the measured vessel diameter reflected the true size of the vessel.

2.3.4. Model-based vessel diameter calculation

The two parametric models were fitted (Fig. 1a) to the one-dimensional intensity profiles from the previous step, using nonlinear least-squares regression in MATLAB.

Standard Gaussian function:

$$f(x) = A \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) + C \quad (1)$$

where A is the peak amplitude, μ is the centre position, σ is the standard deviation, and C is a constant baseline offset.

Generalised-Gaussian (GG) function:

$$f(x) = A \exp\left(-\left(\frac{|x - \mu|}{\alpha}\right)^\beta\right) + C \quad (2)$$

where A is the peak amplitude, μ is the centre position, α is the scale parameter, β is the shape parameter, and C is a constant baseline offset. Given the imaging system configuration, all vessels are expected to produce at minimum a Gaussian intensity profile in the B-scan cross-section, and so profiles sharper than Gaussian profile ($\beta < 2$) are therefore not technically meaningful. The shape parameter β was hence constrained to $\beta \geq 2$ during fitting. With this constraint, the GG accommodated to profiles ranging from Gaussian ($\beta = 2$, where $\sigma = \alpha/\sqrt{2}$) to increasingly flat-topped ($\beta > 2$).

Two standard width metrics were calculated from the fitted models: the full width at half maximum (FWHM) and the $1/e^2$ width. For the standard Gaussian fit, the $1/e^2$ width was calculated as:

$$w_{1/e^2, G} = 4\sigma \quad (3)$$

The FWHM for a Gaussian is related to the $1/e^2$ width by a fixed proportionality:

$$w_{fwhm, G} = 2\sigma\sqrt{2\ln 2} = \frac{\sqrt{2\ln 2}}{2} w_{1/e^2, G} \quad (4)$$

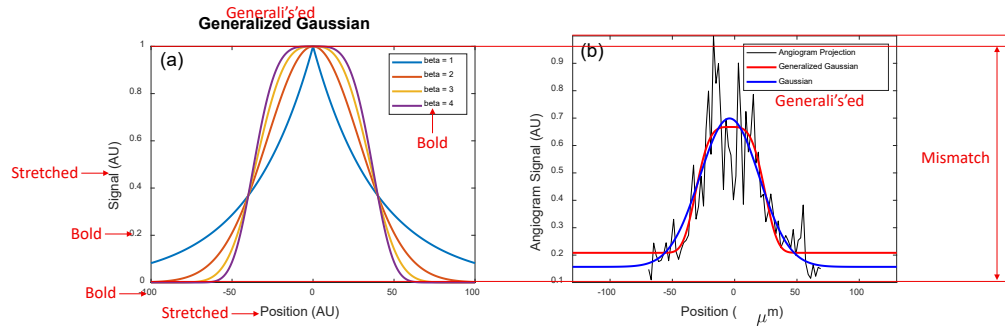


Fig. 1. (a) GG function for shape parameters $\beta = 1, 2, 3$, and 4. (b) Example vessel intensity profile (black) with standard Gaussian (blue) and GG (red) fits.

and therefore only the $1/e^2$ width was calculated directly from the fit, with the FWHM obtainable via this known relationship.

For the GG fit, no such linear relationship exists between the FWHM and $1/e^2$ width, since both depend on the shape parameter β . Both metrics were therefore calculated independently as:

$$w_{fwhm, GG} = 2\alpha(\ln 2)^{1/\beta} \quad (5)$$

$$w_{1/e^2, GG} = 2\alpha \cdot 2^{1/\beta} \quad (6)$$

2.3.5. Binary mask based vessel diameter calculation

For comparison with OCTA processing pipelines that rely on image binarisation, vessel diameters were also measured from binary masks generated by global thresholding. A 3D averaging filter with a kernel size of $50 \times 50 \times 10$ voxels (lateral $x \times$ axial $z \times$ temporal t) was applied to the DyC-OCT angiogram volumes to reduce noise and minimize the effects of outliers. To produce a binarised black and white (BW) mask, the maximum angiogram intensity within these smoothed volumes were calculated, and five different threshold values were defined as fixed percentages of these maxima for MIP and MeIP. For MIP, the thresholds were defined as BW-1 = 40%, BW-2 = 50%, BW-3 = 60%, BW-4 = 70%, and BW-5 = 80%. While for MeIP, the thresholds were defined as BW-1 = 07%, BW-2 = 08%, BW-3 = 09%, BW-4 = 10%, and BW-5 = 11%. These thresholds were selected empirically to cover the usable range of each projection. The upper and lower bounds were selected based on the point at which vessels began to be lost, and at which background noise and tissue signal were increasingly included, respectively. Each threshold was applied to the unsmoothed en face projection (MIP and MeIP) to generate a binary mask, and the vessel diameter was measured as the number of pixels above the threshold across the lateral vessel cross-section.

2.3.6. Statistical analysis

To evaluate the accuracy and precision of all diameter measurement approaches, the calculated diameter values were plotted against the ground truth for each vessel across three conditions: pre-contrast, peak contrast, and post-contrast. A linear regression was fitted to each condition, yielding the slope (m) and coefficient of determination (R^2). A slope of $m = 1$ indicated perfect agreement with the ground truth, and R^2 quantified the goodness of fit of the linear model. Additionally, the coefficient of variation (CoV) was computed for each method to assess measurement precision (repeatability). For the model-based methods, six combinations were evaluated: two projection methods (MIP and MeIP) by three width metrics ($w_{1/e^2, G}$, $FWHM_{GG}$,

208 and $w_{1/e^2, GG}$). For the binary mask based method, diameter was measured at five threshold levels
 209 (BW-1 through BW-5) for both MIP and MeIP. For the functional flicker stimulation experiment,
 210 the Pearson correlation coefficient (ρ) between measured diameter values and angiogram intensity
 211 was computed for each method and vessel, to assess whether the diameter measurements were
 212 affected by changes in angiogram intensity. Measurements with $p < 0.05$ were considered
 213 statistically significant. A paired t-test was used to evaluate whether vessel diameters differed
 214 significantly between the pre- and post-stimulation windows.

215 3. Results

216 3.1. Qualitative assessment of vessel underestimation

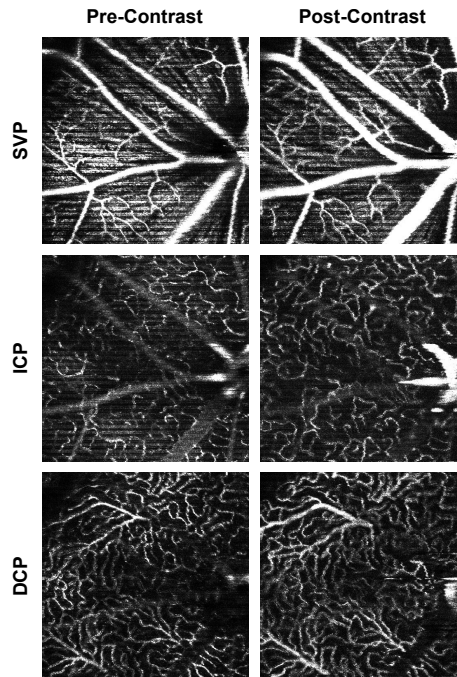


Fig. 2. OCTA MIP en face of the SVP, ICP, and DCP before (left) and after (right) the administration of Intralipid 20%. Vessels appear visibly wider and brighter post-contrast across all layers, most prominently in the larger vessels.

217 Figure 2 shows OCTA MIP enface images of the superficial vascular plexus (SVP), intermediate
 218 capillary plexus (ICP), and deep capillary plexus (DCP) before and after contrast injection. Across
 219 all three layers, vessels appeared visibly wider and brighter post-contrast injection, consistent
 220 with previous literature [8]. This effect was most pronounced in the larger vessels, especially in
 221 the SVP, where the major retinal arteries and veins appeared noticeably thicker after contrast
 222 administration. In the ICP and DCP, the capillary networks also showed improved visibility,
 223 though the effect was less dramatic.

224 DyC OCTA projections helped visualise the temporal changes in vessel diameter pre- and
 225 post-contrast injection (Fig. 3). Pre-contrast injection, the vessel cross-sections appeared narrower
 226 than in the contrast-enhanced phase (Fig. 3a), which could be due to lateral edge signal suppression
 227 by the RBC orientation artefact [4]. At peak contrast, the Intralipid filled the vascular lumen
 228 completely. The vessels appeared slightly wider at peak than post-contrast, and we think this
 229 could be due to the dynamic range of the image rather than a true diameter change (Fig. 3b). In
 230 the post-contrast phase, the signal stabilised as the initial bolus cleared, and the vessel boundaries

231 remained more visible than before injection (Fig. 3c). The MIP and the corresponding binary
 232 mask over time (Figs. 3d–e) showed the observed temporal changes in vessel diameter across the
 233 three phases. The mean OCTA intensity (Fig. 3f) showed a sharp increase post-contrast injection,
 234 followed by a decrease to a new steady state that remained elevated above the pre-contrast
 235 baseline.

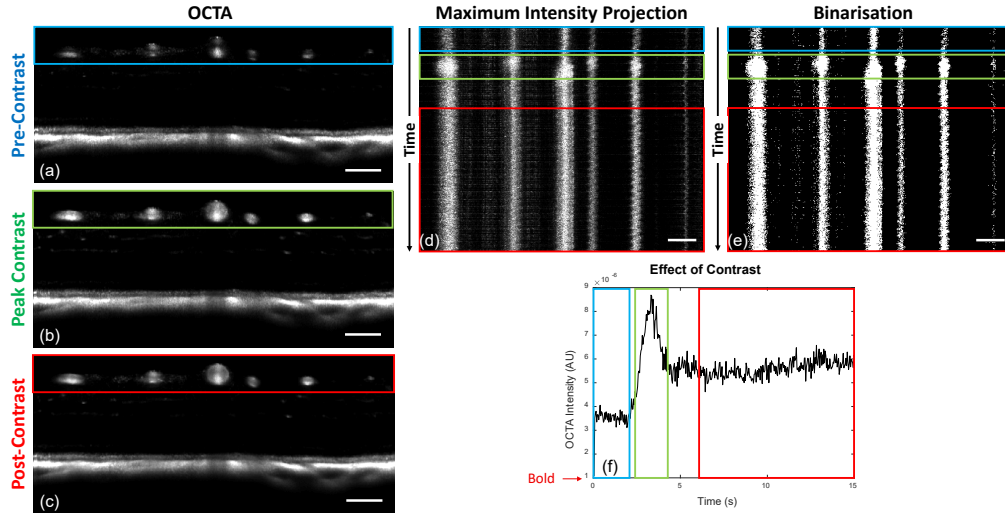


Fig. 3. DyC-OCT B-scan angiograms at (a) pre-contrast, (b) peak contrast, and (c) post-contrast timepoints. (d) MIP and (e) binarisation of the vessel cross-sections over time, with coloured boxes indicating the pre-contrast (blue), peak-contrast (green), and post-contrast (red) windows. (f) Mean OCTA intensity over time, showing the temporal intensity dynamics. All scale bars are 100 μm .

236 3.2. Accuracy and precision of diameter measurements

237 The accuracy (m , R^2) and precision (CoV) of all diameter measurement methods are summarised
 238 in Table 1. All model-based methods underestimated vessel diameter in pre-contrast data, with
 239 slopes ranging from 0.42 to 0.73 (Table 1). Contrast enhancement improved accuracy across
 240 all metrics, though the degree of improvement varied. The $w_{fwhm, GG}$ metric remained well
 241 below unity even post-contrast. The $w_{1/e^2, G}$ with MIP slightly overestimated at both peak and
 242 post-contrast ($m = 1.15$ and $m = 1.12$), while MeIP yielded values closer to unity ($m = 1.04$
 243 in both conditions). The $w_{1/e^2, GG}$ with MeIP provided the best overall agreement, achieving a
 244 slope closest to unity with $m = 1.001$ ($R^2 = 0.97$) post-contrast, which was the closest to the
 245 identity line of all tested model-based methods. **MeIP generally gave slopes closer to unity than**
 246 **MIP, but the linearity showed the opposite trend before contrast. Pre-contrast R^2 was higher for**
 247 **MIP across all three metrics, and MeIP only surpassed MIP post-contrast (Table 1, Fig. 4).**

248 For binary mask based methods, the slope decreased with increasing threshold across all
 249 contrast conditions and projections (Fig. 5, Fig. 6) (Table 1). This degradation was steeper for
 250 MIP than for MeIP. MIP slopes dropped from near unity at BW-1 to well below it at BW-5,
 251 whereas MeIP slopes declined more gradually and remained closer to unity across the threshold
 252 range. MeIP slopes were consistently higher than MIP at every threshold and contrast condition,
 253 though at lower thresholds and at peak and post-contrast, MeIP tended to slightly overestimate
 254 vessel diameter. **The R^2 values showed opposite trends for the two projections. For MIP, R^2**
 255 **decreased with increasing threshold across all contrast conditions, most strongly post-contrast.**

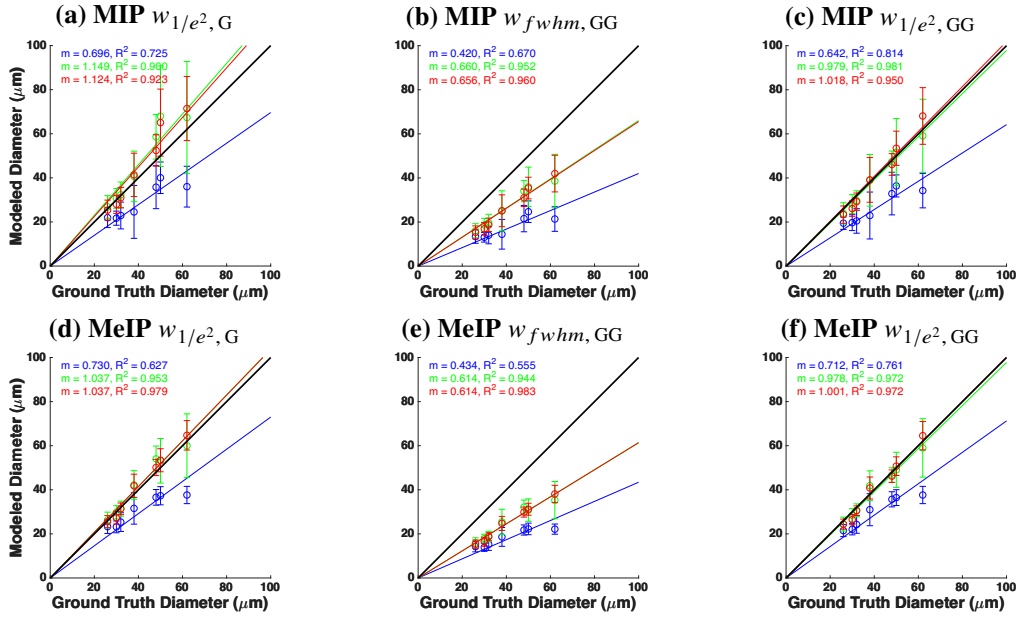


Fig. 4. Modelled versus ground truth vessel diameter using MIP (top row, a–c) and MeIP (bottom row, d–f). Columns correspond to $w_{1/e^2, G}$ (a, d), $w_{fwhm, GG}$ (b, e), and $w_{1/e^2, GG}$ (c, f). Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions. The black line indicates the identity line ($m = 1$).

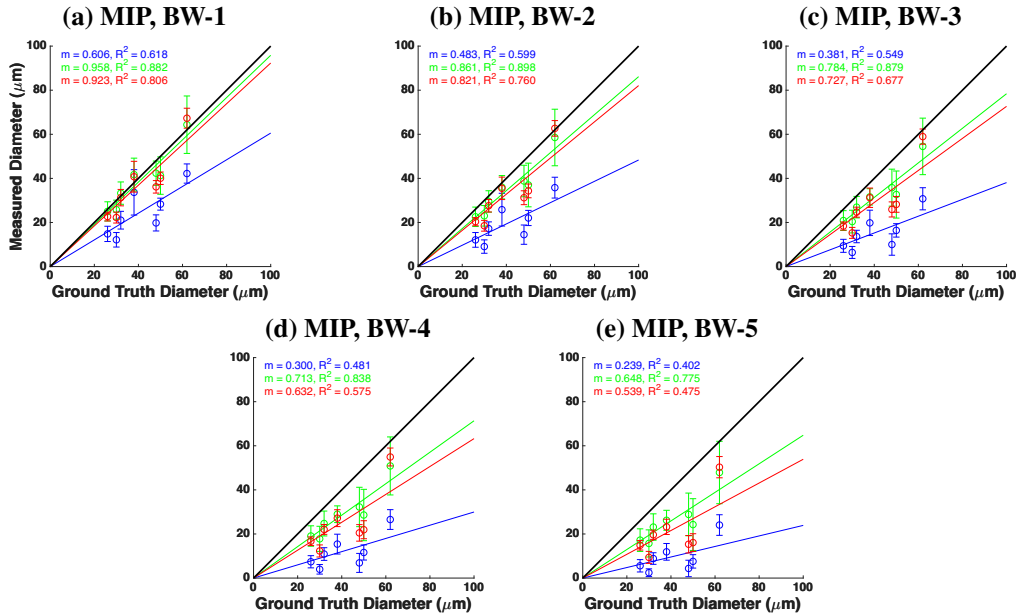


Fig. 5. Vessel diameter measured from binarised masks versus ground truth for MIP at five threshold levels: (a) BW-1 through (e) BW-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

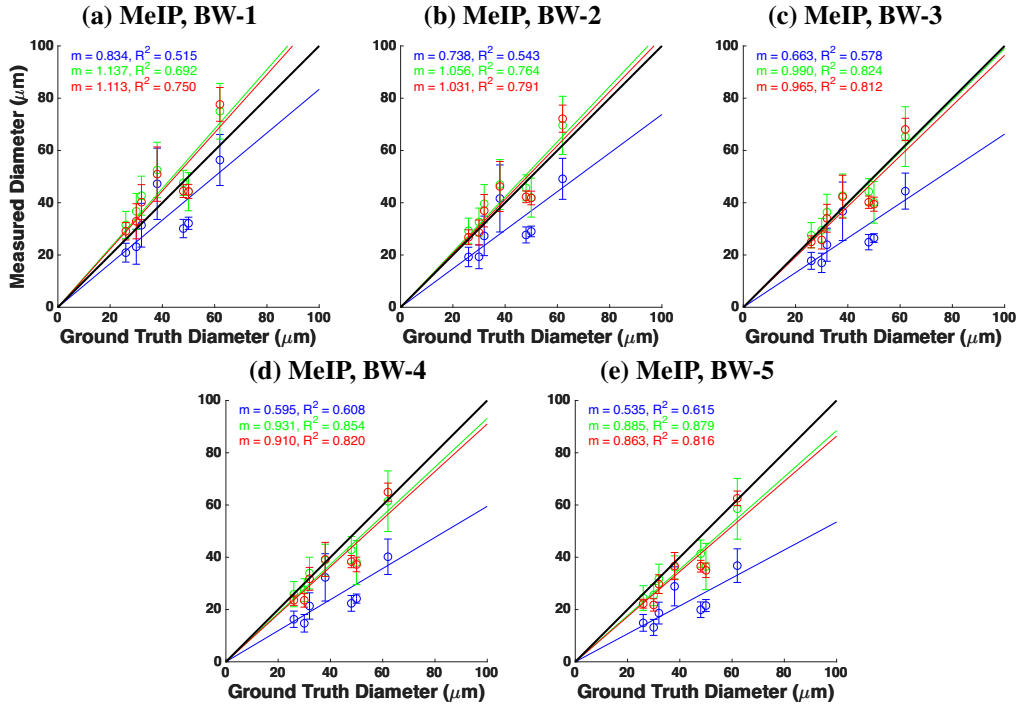


Fig. 6. Vessel diameter measured from binarised masks versus ground truth for MeIP at five threshold levels: (a) BW-1 through (e) BW-5. Linear regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions.

For MeIP, R^2 instead increased with threshold, most evident at peak contrast. Overall, MeIP provided more accurate binary mask based diameter measurements than MIP at every threshold, and was also more linear at the higher thresholds, though MIP provided higher R^2 at the two lowest thresholds (BW-1 and BW-2).

The precision of each method was assessed via the CoV (Table 1, Fig. 7). For MIP, the binary mask based CoV increased substantially with threshold, whereas the model-based CoV remained stable across all contrast conditions. For MeIP, the binary mask based CoV was notably more stable across thresholds than for MIP, and the model-based CoV values were lower overall, though they increased slightly at peak contrast before decreasing post-contrast. Across both methods, MeIP yielded lower CoV values than MIP pre-contrast and more stable values post-contrast. Among the model-based metrics, $w_{fwhm, GG}$ consistently exhibited the weakest accuracy in all three contrast conditions, though its precision remained comparable to the other model metrics. Overall, model-based methods outperformed binary mask based methods in both accuracy and precision, except at lower thresholds for MIP and higher thresholds for MeIP, where precision is comparable to the model-based methods, though accuracy is lower. MeIP yielded superior performance compared to MIP across all methods.

3.3. Functional flicker stimulation

To assess whether angiogram intensity itself biases the diameter measurements, we examined the intensity time series alongside the diameter time series. The model-based diameter time series (Figs. 8a,b) showed increase in diameter post-stimulation for most of the vessels. The binary mask based diameter time series (BW-3; Fig. 8c) showed a mixed trend, where most

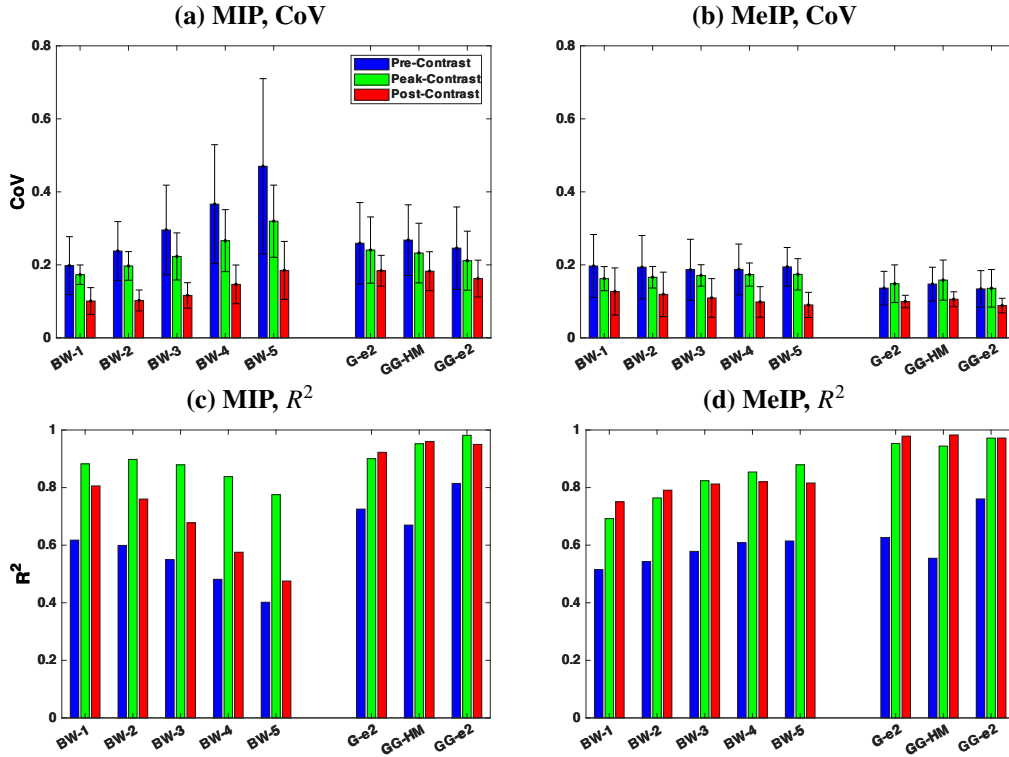


Fig. 7. Summary of accuracy and precision across all diameter measurement methods. (a, b) Coefficient of variation (CoV) and (c, d) coefficient of determination (R^2) for MIP (a, c) and MeIP (b, d), respectively. Bars are grouped by method (BW-1 through BW-5, G-e², GG-HM, GG-e²) and colour-coded by contrast conditions.

Table 1. Summary of slope (m), coefficient of determination (R^2), and coefficient of variation (CoV) for all diameter measurement methods across pre-contrast, peak contrast, and post-contrast conditions. Methods are grouped as binary mask based and model-based. Bold values indicate slope agreement within 10% of unity ($0.9 \leq m \leq 1.1$), strong linearity ($R^2 \geq 0.90$), or high precision (CoV ≤ 0.10).

BW-1			BW-2		BW-3		BW-4		BW-5		$w_{1/e^2, G}$		$w_{fwhm, GG}$		$w_{1/e^2, GG}$	
MIP	MeIP		MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP	MIP	MeIP
<i>Slope (m)</i>																
Pre	0.61	0.83	0.48	0.74	0.38	0.66	0.30	0.60	0.24	0.54	0.70	0.73	0.42	0.43	0.64	0.71
Peak	0.96	1.14	0.86	1.06	0.78	0.99	0.71	0.93	0.65	0.89	1.15	1.04	0.66	0.61	0.98	0.98
Post	0.92	1.11	0.82	1.03	0.73	0.96	0.63	0.91	0.54	0.86	1.12	1.04	0.66	0.61	1.02	1.00
<i>Coefficient of determination (R^2)</i>																
Pre	0.62	0.51	0.60	0.54	0.55	0.58	0.48	0.61	0.40	0.62	0.73	0.63	0.67	0.55	0.81	0.76
Peak	0.88	0.69	0.90	0.76	0.88	0.82	0.84	0.85	0.78	0.88	0.90	0.95	0.95	0.94	0.98	0.97
Post	0.81	0.75	0.76	0.79	0.68	0.81	0.58	0.82	0.48	0.82	0.92	0.98	0.96	0.98	0.95	0.97
<i>Coefficient of variation (CoV)</i>																
Pre	0.20	0.20	0.24	0.19	0.30	0.19	0.37	0.19	0.47	0.19	0.26	0.14	0.27	0.15	0.25	0.13
Peak	0.17	0.16	0.20	0.17	0.22	0.17	0.27	0.17	0.32	0.17	0.24	0.15	0.23	0.16	0.21	0.14
Post	0.10	0.13	0.10	0.12	0.12	0.11	0.15	0.10	0.18	0.09	0.18	0.10	0.18	0.11	0.16	0.09

277 vessels showed a decrease in diameter post-stimulation. Bar plots of the mean diameter pre-
 278 and post-stimulation (Figs. 8d–f) confirmed this pattern. For the Gaussian model, four of five
 279 vessels showed significant diameter increases ($p < 0.01$), and the GG model showed significant
 280 increases in four of five vessels (Table 2). In contrast, the binary mask based diameter showed
 281 significant decreases in four of five vessels.

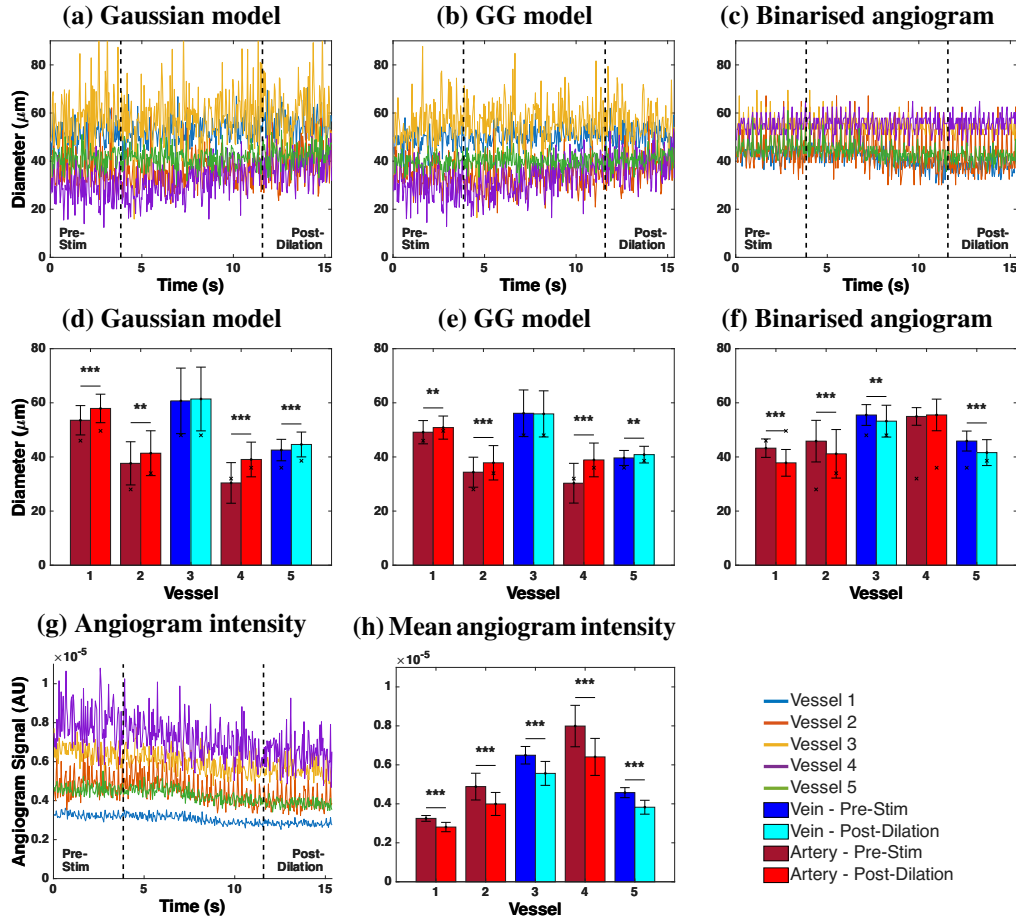


Fig. 8. Functional flicker stimulation results. (a–c) Vessel diameter time series pre- and post-stimulation for (a) Gaussian model, (b) Generalised Gaussian model, and (c) binarised angiogram, across five vessels. Vertical dashed lines indicate stimulation onset and offset. (d–f) Mean diameter per vessel before (dark bars) and after (light bars) stimulation for each method. Veins are shown in blue/cyan and arteries in dark/light red. Significance levels: ** $p < 0.01$, *** $p < 0.001$. (g) Angiogram intensity time series and (h) mean angiogram intensity per vessel pre- and post-stimulation.

282 The angiogram intensity time series (Fig. 8g) showed a visible decline in angiogram intensity
 283 throughout the duration of the DyC-OCT scan across all five vessels. The mean intensity
 284 comparison pre- and post-stimulation (Fig. 8h) confirmed that the decrease was significant for
 285 all vessels, ranging from -13.6% to -19.8% (Table 2). The binary mask based method showed
 286 uniformly positive per-vessel correlations, with an overall correlation of $\rho = 0.526$ ($p < 0.0005$).
 287 The Gaussian and GG models yielded overall correlations of $\rho = -0.101$ and $\rho = -0.098$,
 288 respectively.

Table 2. Flicker stimulation results. Δd : diameter change (%); ρ : Correlation between diameter and angiogram intensity; ΔI : intensity change (%). Statistically significant values are bold (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

	Gaussian		GG		Binarised		Intensity
	Δd (%)	ρ	Δd (%)	ρ	Δd (%)	ρ	
Vessel 1	+8.2***	-0.07	+3.5**	0.03	-12.5***	0.59***	-13.6***
Vessel 2	+9.9**	0.23*	+10.1***	0.07	-10.2***	0.80***	-18.2***
Vessel 3	+1.2	-0.28**	-0.4	-0.28**	-4.1**	0.27**	-14.3***
Vessel 4	+28.5***	-0.15	+28.4***	-0.14	+1.0	0.46***	-19.8***
Vessel 5	+4.9***	-0.24*	+3.1**	-0.18	-9.3***	0.52***	-16.3***
Overall ρ		-0.101		-0.098		0.526	

4. Discussion

4.1. Model-based vs binary mask based diameter measurement

The model-based and binary mask based diameter methods define vessel boundary in fundamentally different ways. Binarisation of the en face maps discards the spatial intensity distribution and reduces each pixel to a binary decision. The measured diameter directly depends on the chosen threshold value and on angiogram intensity, both of which may vary across imaging sessions, contrast conditions, and even between vessels within the same scan. Model-based methods fit a parametric model function to the maximum or mean intensity profiles, and calculate the diameter from the fitted parameters. As the fit captures the shape of the profile rather than the absolute intensity, it is inherently less sensitive to changes in angiogram signal and does not require a threshold at all.

This difference explains the trends observed in Fig. 7 and Table 1. Model-based methods kept slopes closer to the ground truth and maintained stronger linearity across all contrast conditions, whereas for the binary mask based methods both the slope and the linearity degraded progressively with increasing threshold. This distinction matters because a slope that is systematically low but linear can still be corrected, whereas poor linearity cannot. Raising the threshold could therefore remove a vessel dependent fraction of the cross-section, and so the measured width scatters more widely about its mean causing poor linearity. Pre-contrast, the hourglass artefact caused by RBC orientation within vessel lumen suppresses edge signals, causing all methods to underestimate diameter [4]. However, model-based methods still maintained a linear relationship with the ground truth, producing systematic, predictable underestimation. Binary mask based methods on the other hand showed weaker linearity in most conditions, potentially because the suppressed edge signals did not consistently exceed the binarisation threshold, depending on vessel size and local intensity.

The generalised Gaussian (GG) model outperformed the standard Gaussian because the intensity profiles of the vessels are usually not purely Gaussian (Fig. 1b). The standard Gaussian cannot capture relatively flat-topped angiogram intensity profiles and tend to fit a narrower curve, whereas the GG model accommodates through the shape parameter β , which ranges from Gaussian ($\beta = 2$) to increasingly flat-topped profiles ($\beta > 2$). The flat-topped shapes arise because the measured profile is effectively the vessel's reflectivity profile convolved with the system point spread function. Because the GG reduces to a standard Gaussian when $\beta = 2$ and departs from it only when the data support a flatter profile, a correctly applied GG fit should perform at least as well as the Gaussian, at the cost of additional computation. (*new para*) Among the width metrics, FWHM_{GG} consistently showed the weakest accuracy, although its linearity remained high post-contrast ($R^2 = 0.96$ for MIP and 0.98 for MeIP), indicating that

the underestimation is systematic and therefore correctable. Near the half-maximum the profile shape changes rapidly and varies between vessels, from Gaussian to flat-topped, so the width measured at that level is more variable. The $1/e^2$ width captures more of the transition region and so may have provided a more accurate measurement of the vessel diameter.

4.2. Maximum vs mean intensity projection

Across nearly all methods and conditions, MeIP outperformed MIP in both accuracy and precision (Table 1). Linearity was the exception, with pre-contrast MIP producing higher R^2 than MeIP for all model-based metrics, and at the two lowest binarisation thresholds, where MeIP outperformed MIP only after contrast enhancement. This finding is noteworthy because MIP has been shown to produce higher signal-to-noise ratio (SNR) and contrast in en face OCTA images [13], and is the default in many commercial OCTA systems and analysis pipelines [12, 14]. However, we observed that higher SNR did not necessarily translate to more accurate diameter measurements. The MIP selects the single highest intensity value along each A-scan within the vessel ROI, which makes it inherently sensitive to outliers. MeIP averages all intensity values along each A-scan, which smooths out intensity fluctuations and provides a more consistent profile. This averaging could be beneficial for vessel diameter measurement, as the goal is to capture the typical vessel boundary rather than extreme signal events.

The difference between MIP and MeIP was especially apparent in the binary mask based results (Table 1). For MeIP, the slope declined more gradually and remained at 0.86 even at BW-5 post-contrast. Whereas for MIP, the slope decreased steeply with increasing threshold, dropping from 0.92 at BW-1 to 0.54 at BW-5 post-contrast. These findings suggest that MeIP produces a more uniform intensity distribution across the vessel profile, making the binarisation result less sensitive to the exact threshold value. The CoV data (Table 1) reinforced this as the MIP-based binary mask CoV increased substantially with threshold, whereas the MeIP-based CoV remained relatively stable. This behaviour follows from the shape of the two projected profiles. Because the MIP retains only the brightest pixel along each A-scan, the resulting profile is sharply peaked with steep rises and falls, and the peak intensity could vary considerably between vessels depending on the local signal. Unlike MIP, MeIP averages along each A-scan and produces a broader, smoother profile. As a result, thresholding produces a comparable reduction in measured width across vessels, leading to a largely unchanged CoV despite increasing threshold. For model-based methods, MeIP also showed better performance. The $w_{1/e^2, G}$ using MIP slightly overestimated vessel diameter post-contrast with $m = 1.12$, while MeIP reduced this to $m = 1.04$, a much closer agreement with the ground truth. These results suggest that while MIP may be preferred for visualisation purposes where contrast and SNR are important [13], MeIP should be the preferred projection method for quantitative diameter measurements.

4.3. Influence of angiogram intensity on diameter measurements

The functional flicker stimulation experiment provided a direct demonstration of how changes in angiogram signal intensity can affect diameter measurements based on the method used. During the DyC-OCT acquisition, the angiogram intensity declined significantly across all five vessels (Fig. 8g,h), with decreases ranging from -14.1% to -19.9% (Table 2). We are not sure about the cause of this decline, but the inspection of the volumetric data ruled out the most important confounds. There was no out-of-plane motion or breathing artefact, and the image quality was consistent throughout the scan. We suspect a combination of rapid corneal drying and/or cataract formation over the course of the scan could have caused the decline in intensity. More importantly, the cause of the decline is not essential to our argument for this study, what matters is that the intensity decline biases the binary mask-based diameter measurement while the model-based diameter remain unaffected. The results demonstrated that the model-based diameter calculation provided more accurate and repeatable measurements, and also recovered

372 physiologically meaningful vascular responses that binary mask based methods misrepresented.

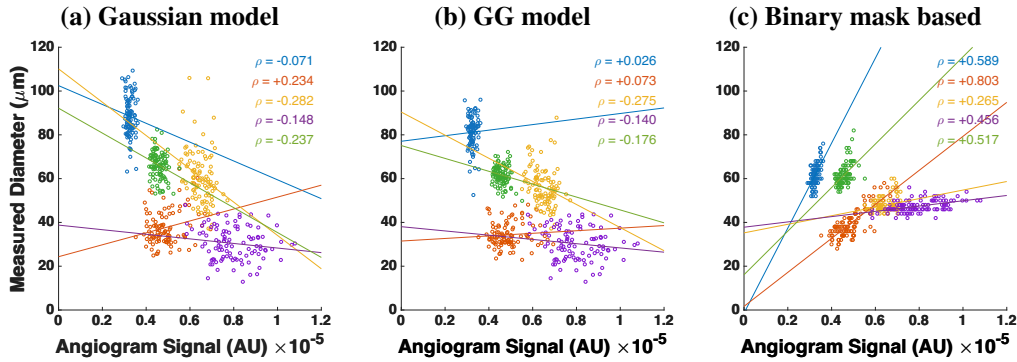


Fig. 9. Measured vessel diameter as a function of angiogram signal intensity during functional flicker stimulation for (a) Gaussian model, (b) Generalised Gaussian model, and (c) binary mask based method. Each colour represents one of five vessels. ρ denotes the Pearson correlation coefficient.

373 The model-based methods correctly detected this expected vasodilation, with the Gaussian
 374 model showing significant diameter increases in four of five vessels (up to +28.5%) and the
 375 GG model showing significant increases in three of five vessels (Table 2). The binary mask
 376 based method, however, reported significant diameter decreases in four of five vessels, suggesting
 377 vasoconstriction which is a paradoxical result. The reason behind this behaviour could be
 378 understood by observing how each method responds to declining signal intensity (Fig. 9). For
 379 the binary mask based method, the threshold is fixed and diameter is determined by the number
 380 of pixels that exceed this threshold. When the overall angiogram intensity drops, fewer pixels at
 381 the vessel edges exceed the threshold, and the vessel appears to shrink even if the physical vessel
 382 diameter has increased. This is confirmed by the strong positive correlation between measured
 383 diameter and angiogram intensity for the binary mask based method ($\rho = 0.523$, $p < 0.0005$). In
 384 contrast, the model-based methods showed weak overall correlations with intensity ($\rho = -0.101$
 385 for Gaussian, $\rho = -0.098$ for GG), indicating that the fitted diameter was effectively independent
 386 of the signal intensity. Among the methods, the GG model also produced the tightest error
 387 bars across the five vessels in the flicker experiment (Fig. 8d–f), consistent with its low CoV
 388 post-contrast (Table 1), indicating that it is the most repeatable of the metrics tested.

389 These findings have important implications for longitudinal studies and any experiment where
 390 angiogram intensity may vary between or within imaging sessions. In longitudinal mouse eye
 391 imaging, the angiogram signal can vary between and within sessions as anaesthesia alters retinal
 392 perfusion and ocular physiology [16, 17], with additional factors such as corneal drying and
 393 cataract formation further reducing OCT intensity over time [18, 19] if artificial tear drops are
 394 not applied during the imaging. If a binary mask based method is used under such conditions,
 395 observed changes in vessel diameter may reflect intensity fluctuations rather than true vascular
 396 changes. In contrast, model-based methods may be less sensitive to these intensity variations,
 397 as they characterise the shape of the intensity profile rather than relying primarily on absolute
 398 intensity values to define vessel boundaries.

399 4.4. Threshold selection

400 The results presented here used a basic, global threshold for the binarisation of the angiogram
 401 data. There are numerous other approaches for binarising the angiogram, which will most
 402 likely each behave differently. For example, angiogram signals can be enhanced with vesselness
 403 filters [20] and dynamic thresholds can be used to binarise images with varying SNR. When

applying a vesselness filter, it is possible to enhance the shape of the angiogram and omit noise, but it is also possible for such enhancement to overcompensate the signal, leading to larger than expected vessel appearance. Dynamic thresholding can also yield a more uniform angiogram appearance, but similarly risks over- or under-compensating vessels given that threshold level directly affects the measured diameter. Further analysis of these other methods would be required to determine their efficacy in returning accurate and precise diameter values.

The key observation from the present work is that any thresholding approach, regardless of how the threshold is determined, shares the same fundamental limitation that the measured diameter depends on the absolute signal intensity at the vessel edges. As demonstrated by the flicker stimulation experiment, this intensity dependence can introduce artefactual diameter changes when the angiogram signal varies over time or between sessions. A more adaptive thresholding strategy may reduce variability across different regions of a single image, but it would not eliminate the sensitivity to temporal intensity fluctuations that we observed. The model-based approach avoids this limitation entirely by deriving diameter from the shape of the intensity profile rather than from an intensity cut-off, making it a fundamentally different and more robust alternative for quantitative diameter analysis.

4.5. Best choices under different conditions

Under the conditions where a contrast agent is available and accuracy is the primary goal, the $w_{1/e^2, GG}$ metric using MeIP provides the best results post-contrast. This combination achieved a slope of $m = 1.001$ with $R^2 = 0.97$ and a CoV of 0.09, yielding the best accuracy, linearity, and precision of all tested methods. If only pre-contrast data is available, the same metric still provides the best linearity ($R^2 = 0.88$) among all methods, though the slope of $m = 0.71$ indicates that diameter could be underestimated by approximately 30%. In this case, a correction factor could be applied if absolute diameter values are needed, though relative comparisons between vessels or timepoints would remain valid without correction.

When only MIP data is available, which is the case for many commercial OCTA machines and existing datasets, the $w_{1/e^2, GG}$ still provides good post-contrast accuracy ($m = 1.02$, $R^2 = 0.93$). For binary mask based analysis of MIP data, only the lowest threshold (BW-1) post-contrast achieved a slope near unity ($m = 0.92$), and the slope degraded rapidly with increasing threshold values. This highlights that if binarisation must be used with MIP data, a lower threshold value is essential, though this may come at the cost of increased noise sensitivity.

If computation time is a limiting factor and model fitting is not practical, MeIP with a moderate threshold (e.g. BW-3) provides a reasonable alternative. At post-contrast, BW-3 using MeIP achieved $m = 0.97$ and $R^2 = 0.80$, which is acceptable for many applications. MeIP-based binary mask based analysis showed more stable CoV values across all thresholds compared to MIP, making the results less sensitive to the exact threshold choice. This stability makes MeIP the preferred projection for binary mask based OCTA.

4.6. Literature comparison

Hormel et al. [13] showed that MIP produces en face OCTA images with higher SNR and contrast compared to MeIP, and concluded that MIP should be preferred for OCTA visualisation. Our results are consistent with this for image quality purposes, but reveal a different picture when it comes to quantitative vessel diameter measurement. We found that MeIP consistently yielded more accurate and precise diameter values than MIP across both model-based and binary mask based approaches. This suggests that the projection method best suited for visualisation is not necessarily the best suited for quantification, and that the choice of projection should depend on the intended analysis.

The challenge of standardising OCTA image analysis has been highlighted by Sampson et al. [14] and Untracht et al. [12], both of whom emphasised the need for transparent, reproducible analysis

452 pipelines. Our findings add to this discussion by showing that the choice of diameter measurement
453 method introduces a source of variability that is at least as significant as the choice of binarisation
454 threshold. The model-based approach presented here offers a path toward more standardised
455 vessel diameter quantification, as it reduces the dependence on selected thresholds and angiogram
456 intensity, and provides measurements that are more robust across imaging conditions.

457 The RBC orientation artefact and its effect on OCTA vessel appearance has been described by
458 Bernucci et al. [4] and Cimalla et al. [3], who used Intralipid contrast to demonstrate that the
459 hourglass pattern in vessel cross-sections is caused by the directional scattering properties of
460 RBCs rather than by true variations in flow. Our work extends this observation by quantifying
461 the degree to which this artefact affects diameter measurements and by proposing model-based
462 fitting as a practical correction strategy. While contrast enhancement with Intralipid is effective
463 for revealing the true vessel diameter, it requires a contrast injection and is not yet applicable in
464 clinical settings. Using a calibrated model-based approach provides a way to partially recover
465 the underestimated diameter from native OCTA data without the need for a contrast agent.

466 Recent work by Son et al. [21] used functional OCT to image flicker-evoked vasodilation in
467 the mouse retina and reported that vasodilation is anisotropic, with larger percent changes in the
468 axial direction ($\sim 18\%$) than the lateral direction ($\sim 11\%$). The authors did not describe how
469 vessel diameters were quantified from their B-scans. It is worth considering whether the lateral
470 diameter measurements may have been affected by the intensity-dependent artefacts observed in
471 the present work. In the lateral direction, the RBC orientation artefact suppresses signal at vessel
472 edges, and depending on how their vessel boundary was defined, this artefact could lead to an
473 underestimation of the lateral diameter. If this were the case, the apparent difference between
474 axial and lateral vasodilation could be influenced by the measurement methodology in addition
475 to the biomechanical anisotropy discussed by the authors. It would be interesting to apply the
476 model-based approach presented here to such data to determine whether their reported anisotropy
477 changes when the lateral diameter is quantified differently.

478 The magnitude of vasodilation detected by our model-based methods ranged from $+1.5\%$
479 to $+28.5\%$ across the five vessels measured (Table 2). The arteries (Vessels 1, 2, and 4)
480 showed the largest diameter increases, with the Gaussian model detecting changes of $+9.2\%$,
481 $+9.6\%$, and $+28.5\%$, respectively, while the veins (Vessels 3 and 5) showed smaller diameter
482 increases of $+1.5\%$ and $+4.9\%$. Table 3 summarises the stimulation protocols and vascular
483 responses reported by previous flicker stimulation studies alongside our own. The listed human
484 studies span several measurement techniques, from fundus photography [22] through retinal
485 vessel analysers [23–25] to Doppler OCT [26] and OCTA [27]. Despite these methodological
486 differences, the reported human arterial responses cluster between $+3\%$ and $+8\%$, and venous
487 responses between $+2\%$ and $+10\%$. Notably, these studies report arterial and venous responses
488 of broadly comparable magnitude, and in some cases the venous response is the larger of the two,
489 for example $+9.8\%$ venous versus $+8.0\%$ arterial in Aschinger et al. [26]. This contrasts with
490 our measurements, in which the arterial response was consistently and substantially larger than
491 the venous response. Whether this asymmetry reflects a genuine feature of the murine flicker
492 response, the small number of vessels sampled, or a greater sensitivity of the model-based fit
493 to the smooth-muscle-driven dilation of arteries remains to be established. Polak et al. [23]
494 additionally showed that the diameter response is relatively stable across flicker frequencies from
495 4 to 40 Hz, suggesting that frequency differences between studies are unlikely to explain the
496 variation.

497 The magnitudes we observed are generally larger than those reported in the human studies,
498 though direct comparison is difficult given the differences in experimental conditions. Species,
499 anaesthesia, stimulation frequency and wavelength, stimulus power and delivery method, stim-
500 ulation duration, dark adaptation period, and imaging modality all vary across the reported
501 studies, and each of these factors could influence the measured vascular response. Among the

studies listed, Son et al. [21] used the most similar setup to ours. They used the same mouse strain (C57BL/6J), the same anaesthetic (ketamine/xylazine), a comparable flicker stimulus (10 Hz, 504 nm), however a different stimulation period of 5 s. A systematic comparison of these parameters could help determine which factors are most influential to the results observed in both studies. Importantly, regardless of the absolute magnitude, the fact that we were able to detect vasodilation at all depended on the measurement method, because binary mask based analysis not only failed to detect vasodilation but reported vasoconstriction which would have led to fundamentally incorrect physiological conclusions.

Previous studies in mice have measured flicker-evoked retinal vessel diameter changes mostly using dynamic vessel analysis (DVA), which is a fundus camera-based technique adapted from human retinal vessel analysers. Albanna et al. [28] demonstrated the feasibility of DVA in the mouse eye using 12.5 Hz flicker at 530 nm but found that quantitative arterial recordings were possible only in a minority of animals, while venous dilation was mostly below 1.5% post-stimulation. Another study by Neumaier et al. [29] used the same setup in wildtype and Cav2.3-deficient mice with overnight dark adaptation and reported median arterial and venous diameter changes of only +1.3% and +1.2%, respectively, in the wildtype group. These values are substantially smaller than those we observed, which could be due to the limited spatial resolution and signal-to-noise ratio of DVA when applied to the smaller mouse eye rather than being a genuine physiological response. The much larger responses we observed may therefore partly result from the higher lateral resolution of OCT combined with the model-based fitting approach, which captures vessel boundary shifts that could fall below the detection limit of DVA. In addition to these diameter-based studies, Rai et al. [30] recently reported that 12 Hz flicker stimulation significantly increased both arterial and venous retinal blood flow in light-adapted C57BL/6J mice, confirming that the mouse retina exhibits a robust neurovascular coupling response. Since blood flow is proportional to the product of vessel cross-sectional area and flow velocity, their findings are consistent with the vasodilation we observed, although a direct quantitative comparison between flow and diameter changes is not straightforward.

Table 3. Comparison of flicker stimulation protocols and reported diameter changes across studies. H = human, M = mouse. DA = dark adaptation. Lat. = lateral, ax. = axial.

Study	Stimulation	Stimulation	Stimulation	DA	Diameter Change (%)	
	Freq. (Hz)	Light Source	Duration (s)	(h)	Artery	Vein
Formaz [22] (H)	10	Xenon arc, $5.4 \log T d^{\S}$	60	—	+4.2	+2.7
Polak [23] (H)	8	$< 550 \text{ nm}$, $\sim 150 \mu\text{W}$, $\text{cm}^{-2\dagger}$	60	—	+3.5	+1.9
Garhofer [24] (H)	8	$< 550 \text{ nm}$, $300 \mu\text{W}$, $\text{cm}^{-2\dagger}$	60	—	+3.8	+2.8
Nagel [25] (H)	12.5	$530 - 600 \text{ nm}$, $\sim 196 \mu\text{W}$, $\text{cm}^{-2\dagger}$	20	—	+6.9	+6.5
Aschinger [26] (H)	12.5	$530 - 600 \text{ nm}$, $\sim 196 \mu\text{W}$, $\text{cm}^{-2\dagger}$	120	—	+8.0	+9.8
Kallab [27] (H)	7	White, $\sim 450 \text{ lux}$	*	—	+3.7	+4.5
Albanna [28] (M)	12.5	530 nm , 30 lux	20	12	— [#]	$< 1.5^{\#}$
Neumaier [29] (M)	12.5	530 nm , 30 lux	20	12	+1.3	+1.2
Son [21] (M)	10	504 nm , $\sim 460 \text{ lux}$	5	1 – 2	lat. +10.8, ax. +18.2	lat. +8.9, ax. +14.3
Present study (M)	10	502 nm , $26.6 \mu\text{W}^{\ddagger}$	Cont.	2	+9.2 to +28.5	+1.5 to +4.9

^{\S}Diameter measured from fundus photographs taken post-stimulation; retinal luminance in trolands.

[#]DVA feasibility study; arterial recordings were not reliably obtainable in the mouse eye.

*Flicker applied during OCTA scan acquisition. [†]Retinal irradiance. [‡]Power at cornea.

529 4.7. Future direction

530 A future direction of this work is to move toward fully automated segmentation and calculation of
531 vessel diameters from en face angiogram maps. While ROIs were manually selected in this work,
532 it is also possible to automatically segment angiographic data using traditional skeletonisation
533 methods, which are commonly used for automated OCTA analysis. From a skeletonised image,
534 vessel angle can be computed for further extraction of vascular cross-sections perpendicular to
535 the vessel axis. These cross-sections could then be processed using the model-based methods
536 described here to enable fully automated analysis of 3D OCTA data sets, rather than the semi-
537 automated approach shown here for 2D OCTA data. This type of model-based approach is
538 particularly important for comparing 3D volumes acquired at different time points in the mouse
539 eye, as there are often intensity losses caused by drying of the eye, cataract formation, etc. As we
540 have shown in this work, even low levels of intensity loss occurring over a short time window
541 can significantly affect the measured vessel diameter when using a binary mask based approach,
542 but not when using the models presented here.

543 Additionally, this work focused on the larger vessels of the superficial vascular plexus.
544 Extending the model-based approach to the intermediate and deep capillary plexuses, where
545 vessels are smaller and signal levels are lower, would be valuable for studying microvascular
546 changes in diseases. For very small vessels approaching the lateral resolution limit of the imaging
547 system, the measured intensity profile is a convolution of the true vessel profile with the point
548 spread function (PSF) of the incident beam on the retina. Since the lateral resolution of OCT is
549 typically worse than the axial resolution and is determined by the beam optics rather than the
550 source bandwidth, this convolution becomes increasingly significant for smaller vessels. If the
551 PSF were known, a deconvolution approach could be used to recover the true vessel profile before
552 model fitting, potentially improving accuracy for capillary-scale vessels. However, characterising
553 the retinal PSF in vivo remains challenging, as it depends on the ocular optics of each individual
554 eye and may vary with retinal depth and eccentricity.

555 References

- 556 1. V. J. Srinivasan, J. Y. Jiang, M. A. Yaseen, *et al.*, “Rapid volumetric angiography of cortical microvasculature with
557 optical coherence tomography,” *Opt. Lett.* **35**, 43–45 (2010).
- 558 2. A. Javed, A. Khanna, E. Palmer, *et al.*, “Optical coherence tomography angiography: A review of the current
559 literature,” *J. Int. Med. Res.* **51**, 03000605231187933 (2023).
- 560 3. P. Cimalla, J. Walther, M. Mittasch, and E. Koch, “Shear flow-induced optical inhomogeneity of blood assessed in
561 vivo and in vitro by spectral domain optical coherence tomography in the 1.3 μm wavelength range,” *J. Biomed. Opt.*
562 **16**, 116020 (2011).
- 563 4. M. T. Bernucci, C. W. Merkle, and V. J. Srinivasan, “Investigation of artifacts in retinal and choroidal OCT
564 angiography with a contrast agent,” *Biomed. Opt. Express* **9**, 1020–1040 (2018).
- 565 5. Y. Pan, J. You, N. D. Volkow, *et al.*, “Ultrasensitive detection of 3D cerebral microvascular network dynamics in
566 vivo,” *NeuroImage* **103**, 492–501 (2014).
- 567 6. C. W. Merkle and V. J. Srinivasan, “Laminar microvascular transit time distribution in the mouse somatosensory
568 cortex revealed by Dynamic Contrast Optical Coherence Tomography,” *NeuroImage* **125**, 350–362 (2016).
- 569 7. S. Fialová, M. Augustin, M. Glösmann, *et al.*, “Polarization properties of single layers in the posterior eyes of
570 mice and rats investigated using high resolution polarization sensitive optical coherence tomography,” *Biomed. Opt.*
571 *Express* **7**, 1479–1495 (2016).
- 572 8. C. W. Merkle, M. Augustin, D. J. Harper, *et al.*, “High-resolution, depth-resolved vascular leakage measurements
573 using contrast-enhanced, correlation-gated optical coherence tomography in mice,” *Biomed. Opt. Express* **12**,
574 1774–1791 (2021).
- 575 9. Y. Fu and K.-W. Yau, “Phototransduction in mouse rods and cones,” *Pflügers Arch. J. Physiol.* **454**, 805–819 (2007).
- 576 10. M. Augustin, S. Fialová, R. Plasenzotti, *et al.*, “Multi-functional OCT image processing for rodent eyes,” in *Optical*
577 *Tomography and Spectroscopy*, (OSA - The Optical Society, Center for Medical Physics and Biomedical Engineering,
578 Medical University of Vienna, Waehringer Guertel 18-20, 4L, Vienna, A-1090, Austria, 2016), p. 3.
- 579 11. D. J. Harper, M. Augustin, A. Lichtenegger, *et al.*, “Retinal analysis of a mouse model of Alzheimer’s disease with
580 multicontrast optical coherence tomography,” *Neurophotonics* **7**, 015006 (2020).
- 581 12. G. R. Untracht, M. S. Durkee, M. Zhao, *et al.*, “Towards standardising retinal oct angiography image analysis with
582 open-source toolbox octava,” *Sci. Reports* **14** (2024).

- 583 13. T. T. Hormel, J. Wang, S. T. Bailey, *et al.*, "Maximum value projection produces better en face oct angiograms than
584 mean value projection," *Biomed. Opt. Express* **9**, 6412 (2018).
- 585 14. D. M. Sampson, A. M. Dubis, F. K. Chen, *et al.*, "Towards standardizing retinal optical coherence tomography
586 angiography: a review," *Light. Sci. & Appl.* **11** (2022).
- 587 15. C. S. Lee, A. J. Tying, Y. Wu, *et al.*, "Generating retinal flow maps from structural optical coherence tomography
588 with artificial intelligence," *Sci. Reports* **9** (2019).
- 589 16. S. Worm, G. Ladurner, L. May, *et al.*, "Retinal OCT parameters and physiological dynamics as an effect of anesthesia
590 in C57BL/6j mice," *Investig. Ophthalmol. & Vis. Sci.* **67**, 50 (2026).
- 591 17. G. Feng, A. Joseph, K. Dholakia, *et al.*, "High-resolution structural and functional retinal imaging in the awake
592 behaving mouse," *Commun. Biol.* **6**, 572 (2023).
- 593 18. D. M. Stein, G. Wollstein, H. Ishikawa, *et al.*, "Effect of corneal drying on optical coherence tomography,"
594 *Ophthalmology* **113**, 985–991 (2006).
- 595 19. G. J. McLellan and C. A. Rasmussen, "Optical coherence tomography for the evaluation of retinal and optic nerve
596 morphology in animal subjects: Practical considerations," *Vet. Ophthalmol.* **15**, 13–28 (2012).
- 597 20. A. F. Frangi, W. J. Niessen, K. L. Vincken, and M. A. Viergever, "Multiscale vessel enhancement filtering," in *Medical
598 Image Computing and Computer-Assisted Intervention — MICCAI'98*, vol. 1496 of *Lecture Notes in Computer
599 Science* W. M. Wells, A. Colchester, and S. Delp, eds. (Springer, Berlin, Heidelberg, 1998), pp. 130–137.
- 600 21. T. Son, G. Ma, and X. Yao, "Functional OCT reveals anisotropic changes of retinal flicker-evoked vasodilation," *Opt.
601 Lett.* **49**, 2121–2124 (2024).
- 602 22. F. Formaz, C. E. Riva, and M. Geiser, "Diffuse luminance flicker increases retinal vessel diameter in humans," *Curr.
603 Eye Res.* **16**, 1252–1257 (1997).
- 604 23. K. Polak, L. Schmetterer, and C. E. Riva, "Influence of flicker frequency on flicker-induced changes of retinal vessel
605 diameter," *Investig. Ophthalmol. & Vis. Sci.* **43**, 2721–2726 (2002).
- 606 24. G. Garhöfer, C. Zawinka, H. Resch, *et al.*, "Diffuse luminance flicker increases blood flow in major retinal arteries
607 and veins," *Vis. Res.* **44**, 833–838 (2004).
- 608 25. E. Nagel and W. Vilser, "Flicker observation light induces diameter response in retinal arterioles: A clinical
609 methodological study," *Der Ophthalmol.* **102**, 305–309 (2005).
- 610 26. G. C. Aschinger, L. Schmetterer, K. Fondi, *et al.*, "Effect of Diffuse Luminance Flicker Light Stimulation on Total
611 Retinal Blood Flow Assessed With Dual-Beam Bidirectional Doppler OCT," *Investig. Ophthalmol. & Vis. Sci.* **58**,
612 1167–1178 (2017).
- 613 27. M. Kallab, N. Hommer, B. Tan, *et al.*, "Plexus-specific effect of flicker-light stimulation on the retinal microvasculature
614 assessed with optical coherence tomography angiography," *Am. J. Physiol. Circ. Physiol.* **320**, H23–H28 (2020).
- 615 28. W. Albanna, K. Kotliar, J. N. Lüke, *et al.*, "Non-invasive evaluation of neurovascular coupling in the murine retina by
616 dynamic retinal vessel analysis," *PLOS ONE* **13**, e0204689 (2018).
- 617 29. F. Neumaier, K. Kotliar, R. H. L. Haeren, *et al.*, "Evaluation of flicker-induced retinal vessel response in cav2.3-deficient
618 mice using dynamic retinal vessel analysis," *Front. Neurol.* **12**, 659890 (2021).
- 619 30. M. Rai, Y. Lakshmanan, K. Y. Choi, and H. H.-I. Chan, "Effects of flickering light stimulation on retinal blood flow
620 and full-field electroretinogram in mice," *Documenta Ophthalmol.* **151**, 205–218 (2025).